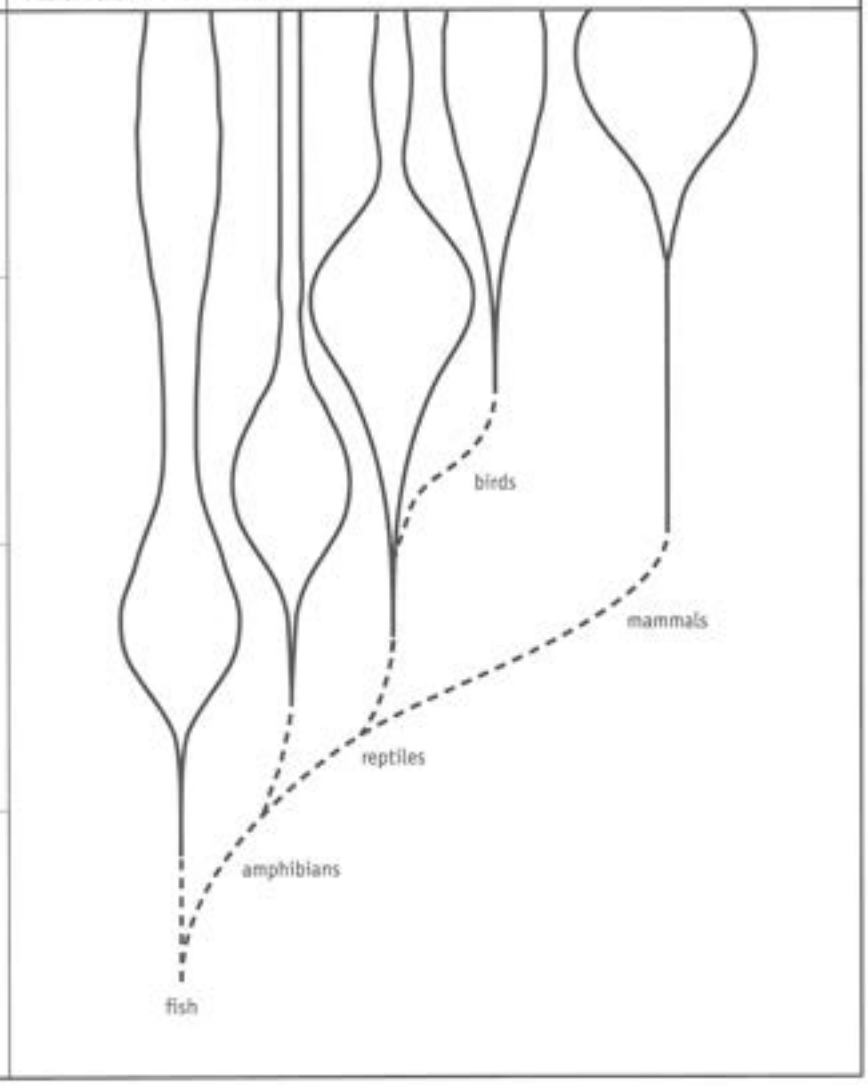
Abundance of fossils

16a

Scientists have been able to work out how many animals of various groups lived in the past. Their findings are based on fossil evidence and are summarised in the table below. Geological time of the earth is divided into eras, which in turn are divided into periods. The table shows the eras only.

Years ago	Era	Abundance of fossils
0	Cainozoic	
65 000 000	Mesozoic	
230 000 000	Paleozoic	
600 000 000	Precambrian	
3 000 000 000		

1 **a** The table above is not to scale, ie the eras have been allocated the same amount of space in the table, even though they are not actually equal lengths of time. Show the eras on the time line below by dividing the line into 500 000 000 year sections. Write the number of '*years ago*' below the time line, then mark the relevant years for each era and label the eras above the time line.

3 billion years ago — today

b Name some difficulties you would have to overcome if you had to redraw the table above to scale.

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Abundance of fossils 16b

Answer the following comprehension questions about the table above.

2 *On the lines*

a When did the Mesozoic era start? _____

b When did the Paleozoic era finish? _____

c What was the most plentiful animal group during the Mesozoic era? _____

d Which is the most abundant vertebrate group during the Cainozoic? _____

3 *Between the lines*

a How long did the Precambrian era last? _____

b Which is the shortest era? _____

c Which is the longest era? _____

d What do the solid lines indicate? _____

e Which was the most abundant animal group about 3000 million years ago? _____

f Are the fish or reptiles more abundant during the Cainozoic era? _____

g Which era is the age of dinosaurs? _____

h Is it true that the age of dinosaurs lasted for about 150 000 000 years? _____

i When did vertebrates originate? _____

4 *Beyond the lines*

a Why should the Precambrian era be so much longer than the other eras? _____

b What do the dotted lines indicate? _____

c What does the dotted line indicate which is branching off the reptiles during the late Paleozoic era? _____

d Why did the abundance of fish decrease during the late Paleozoic era? _____

_____

e There is a theory that the reptiles died back because the earth became suddenly colder. Why didn't the mammals also die back? _____

_____

f Why did the mammals increase in numbers when the reptiles decreased? _____

_____

g Do the fish from the Paleozoic look the same as fish from the Cainozoic? _____

h What is a current example of each vertebrate group? _____

i What is an extinct example of each vertebrate group? _____

j What are the names of five other animal groups? _____

k What is the name of the kingdom and the phylum that the five classes in the table belong to?

_____

l Why did the increase in flowering plants occur at the same time as an increase in birds?

_____

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